

On the Irreducibility of the Coherent Residual Channel: Ablation of Collection and Distribution Layers Extension 3

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Abstract

The Q-LINK architecture utilizes a two-part residual mechanism: a collection layer (U_{coll}) and a distribution layer (U_{dist}). This note formally tests the hypothesis of mechanism ablation to determine the relative contribution of each half. By evaluating the iterated Lie brackets of the ablated circuits, we prove that both the collection-only and distribution-only architectures strictly fracture the Dynamical Lie Algebra (DLA). The generation of the full $\mathfrak{su}(2^{n+1})$ target space requires non-commuting generators on the messenger qubit. We conclude that the residual mechanism is mathematically irreducible; removing either half collapses the structural advantage.

1 The DLA of the Unablated Baseline

The full Q-LINK architecture operates via two interlaced generator sets acting on the extended Hilbert space $\mathcal{H}_{tot} = \mathcal{H}_{data} \otimes \mathcal{H}_m$:

- **Data Operations (U_{opr}):** Contribute generators of the form $I_m \otimes A$, where $A \in \mathfrak{su}(2^n)$.
- **Collection (U_{coll}):** The R_{xx} gates contribute generators proportional to $X_m \otimes X_i$.
- **Distribution (U_{dist}):** The $CNOT_{m \rightarrow i}$ gates (controlled on m) contribute generators proportional to $Z_m \otimes X_i$.

The unablated DLA generates the full messenger Pauli group via the commutator $[X_m \otimes X_i, Z_m \otimes X_j] = -2i(Y_m \otimes X_i X_j)$, ensuring full saturation of $\mathfrak{su}(2^{n+1})$.

2 Ablation 1: Distribution-Only (Algebraic Stagnation)

Suppose we ablate the collection layer, leaving only the U_{opr} and U_{dist} blocks. The available generator set is $\mathcal{G}_{dist} = \{I_m \otimes A, Z_m \otimes B\}$.

We evaluate the algebraic closure under nested commutators:

$$[I_m \otimes A + Z_m \otimes B, I_m \otimes C + Z_m \otimes D] = I_m \otimes [A, C] + I_m \otimes [B, D] + Z_m \otimes ([A, D] + [B, C]) \quad (1)$$

The Lie brackets strictly close within the $\{I_m, Z_m\}$ subspace.

Theorem of Stagnation: The DLA of the distribution-only circuit cannot generate X_m or Y_m . Because the messenger qubit is initialized in $|0\rangle_m$ (an eigenstate of Z_m), it undergoes strictly zero population transfer. The messenger state remains stationary at $|0\rangle_m \langle 0|_m$ for the entire sequence. Consequently, every $CNOT_{m \rightarrow i}$ gate evaluates trivially as the identity. The distribution-only circuit analytically collapses back into the Vanilla circuit, and all variance advantage evaporates.

3 Ablation 2: Collection-Only (Algebraic Fracturing)

Suppose we ablate the distribution layer, leaving only the U_{coll} and U_{opr} blocks. The available generator set is $\mathcal{G}_{coll} = \{I_m \otimes A, X_m \otimes B\}$.

We evaluate the algebraic closure:

$$[I_m \otimes A + X_m \otimes B, I_m \otimes C + X_m \otimes D] = I_m \otimes [A, C] + I_m \otimes [B, D] + X_m \otimes ([A, D] + [B, C]) \quad (2)$$

The Lie brackets strictly close within the $\{I_m, X_m\}$ subspace.

Theorem of Fracturing: The DLA of the collection-only circuit cannot generate Y_m or Z_m . The algebra fractures into $\mathfrak{g} \cong \mathfrak{su}(2^n) \oplus \mathfrak{su}(2^n)$. While the messenger qubit now undergoes population transfer (unlike the distribution-only ablation), it functions merely as a static phase sink rather than a coherent residual loop. Because the algebra fails to span the full $\mathfrak{su}(2^{n+1})$ geometry, the architecture fails to leverage the volumetric trace discrepancy defined by the Master Formula.

4 Conclusion

The Q-LINK residual pathway is irreducible. The architectural advantage relies fundamentally on the non-commutativity of the collection and distribution layers (i.e., $[X_m, Z_m] \neq 0$). Ablating either half of the mechanism isolates the messenger qubit in a commuting sub-algebra, triggering either a trivial reduction to the Vanilla baseline or a fatal fracture of the Dynamical Lie Algebra.